

First Field Trial of Coherent TDM-PON Up to 300 Gb/s

Yixiao Zhu^(1,4,*), Luyao Huang^(2,4), Yimin Hu⁽¹⁾, Ziheng Zhang⁽¹⁾, Chaofeng Ge⁽³⁾, Dezhi Zhang⁽²⁾, Ming Jiang^(2,*), and Weisheng Hu⁽¹⁾

⁽¹⁾ State Key Laboratory of Photonics and Communications, Shanghai Jiao Tong University, Shanghai 200240, China, *yixiaozhu@sjtu.edu.cn

⁽²⁾ China Telecom Research Institute, State Key Laboratory of Optical Fiber and Cable Manufacture Technology, Beijing, 102209, China, *jming@chinatelecom.cn

⁽³⁾ China Telecom Research Institute, Beijing, 102209, China, ⁽⁴⁾ Equal contributions

Abstract We report the first field trial demonstration of coherent TDM-PON over 57-km deployed urban fiber (~34-dB loss), achieving flexible rates up to 300 Gb/s and fast-convergence DSP with a 30-ns preamble. The power budgets are 42/39 dB for 200-/300-Gb/s data rates. © 2026 The Author(s)

Introduction

The rapid growth of bandwidth-intensive applications, including artificial intelligence (AI), 8K video streaming, and cloud computing, is driving the demand for next-generation passive optical networks (PONs) beyond 50G-PON [1]. Very High Speed PON (VHSP), targeting 200 Gbit/s service rates, is currently under active investigation in ITU-T SG15/Q2 as the next evolutionary step [2]. However, conventional intensity modulation with direct detection (IM/DD) faces fundamental limitations for single-wavelength operation at this rate, primarily due to chromatic dispersion penalties and limited receiver responsivity [3].

Coherent detection has emerged as a promising solution for 200G-PON, offering high receiver sensitivity, improved power budget, and full compensation of chromatic dispersion through post digital signal processing (DSP) [4-5]. By leveraging phase and polarization diversity, coherent optics can achieve the target line rates with 4× higher spectral efficiency compared to IM/DD, thereby relaxing component bandwidth requirements [6].

While laboratory experimental demonstrations have validated the feasibility of coherent PON [7],

their performance under real deployed fiber conditions remains uncertain. Field-installed links introduce practical impairments, such as excess splice loss, connector imperfections, random environmental perturbations, and non-ideal dispersion profiles, which may significantly impact system robustness.

Recently, pioneering field trials have explored 200G-class coherent PON. Yan *et al.* reported a bidirectional TFDM coherent PON with real-time FPGA-based reception over 30 km, achieving a downstream peak rate of 200.5 Gb/s [7]. Kovacs *et al.* demonstrated 200 Gb/s bidirectional transmission over 19-km field-installed fiber using Alamouti-coding-based simplified coherent receivers [8]. However, further investigation is required to assess higher line rates, extended reach, and burst-mode operation under realistic deployment conditions.

In this work, we present the first field trial of coherent TDM-PON over 57-km deployed fiber in urban region, achieving line rates up to 300 Gb/s. Fast-convergence DSP is implemented to support signal recovery with a 30-ns preamble. Flexible-rate transmission from 120 to 300 Gb/s is demonstrated, with power budgets up to 42 dB at

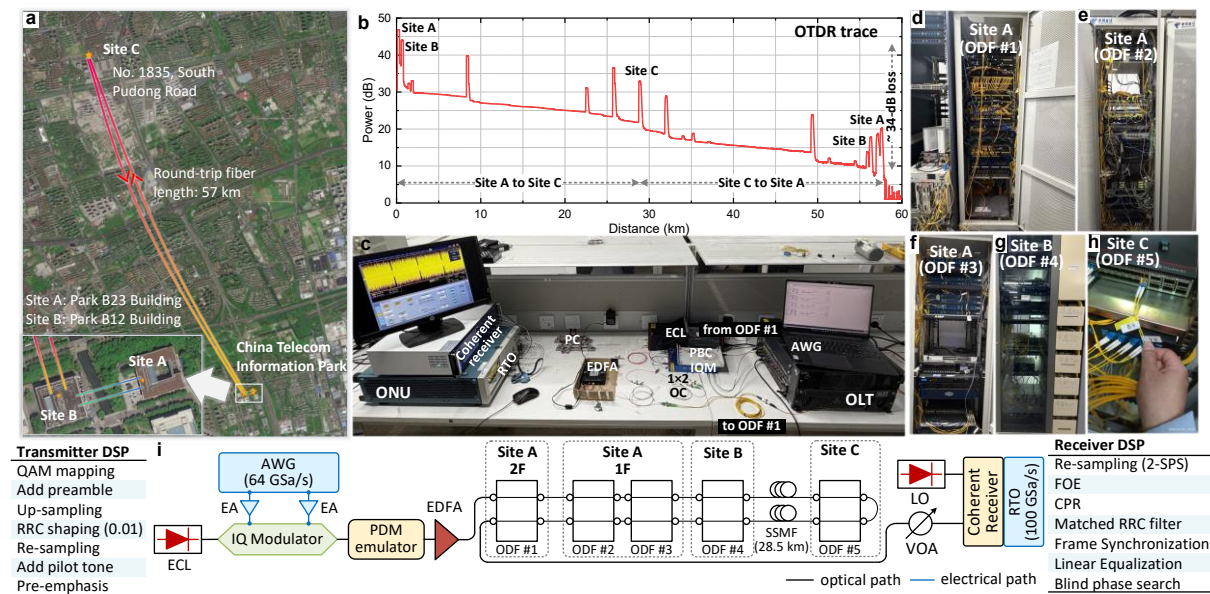


Fig. 1: (a) Geographical map of 57-km field-deployed fiber. (b) OTDR trace of the field-deployed fiber. (c) Photograph of the OLT and ONU. (d)-(h) Photographs of Sites A-C and optical distribution frames (ODFs) #1~#5. (i) Experimental setup and transmitter/receiver DSP blocks. PDM: polarization division multiplexing; FOE: frequency offset estimation; CPR: carrier phase recovery.

200 Gb/s and 39 dB at 300 Gb/s, respectively.

Experimental Setup and DSP

Fig. 1(a) illustrates the field-trial deployment, comprising three sites: (A) B23 Building and (B) B12 Building in China Telecom Information Park, and (C) No. 1835 South Pudong Road. The fiber length is ~ 200 m between Sites A and B, and 28.5 km between Sites B and C, resulting in a total round-trip distance of 57 km in an urban environment. Fig. 1(b) shows the measured optical time-domain reflectometer (OTDR) trace, indicating an overall loss of ~ 34 dB, including fiber, connector, and splice losses. Fig. 1(c) shows the photograph of the experimental setup. Fig. 1(d)-1(h) presents the photographs of the optical distribution frames (ODFs) located at Site A/B/C, respectively.

The detailed experimental setup of the coherent TDM-PON is depicted in Fig. 1(i). At the transmitter, a 100-kHz external cavity laser (ECL, ID Photonics) serves as the optical source. The electrical waveform is generated by a 64-GSa/s arbitrary waveform generator (AWG, Keysight M8195), which drives a 40-GHz IQ modulator biased at the null point. A polarization-division-multiplexing (PDM) emulator is implemented using a polarization-maintaining optical coupler (PM-OC) and a polarization beam combiner (PBC). The dual-polarization signal is amplified to 10 dBm by an erbium-doped fiber amplifier (EDFA) and launched into the field-installed fiber.

At the receiver, a variable optical attenuator (VOA) is used to adjust the received optical

power (ROP). The signal is mixed with a 100-kHz local oscillator (LO) in a 40-GHz coherent receiver with integrated trans-impedance amplifiers (TIAs), and then sampled by a 100-GSa/s real-time oscilloscope (RTO, Tektronix) for offline DSP processing.

The transmitter and receiver DSP blocks are also shown in Fig. 1(i). We use single-carrier modulation. the bit stream is mapped to QPSK, probabilistic-shaped (PS)-16-QAM, or 16-QAM formats for flexible-rate operation. A 1500-symbol preamble is inserted before a 40000-symbol payload. After 2-times up-sampling, root-raised cosine (RRC) filtering with a roll-off factor of 0.01 is applied, followed by re-sampling to generate symbol rates of 30 and 50 GBd. A pilot tone is inserted at the edge of the signal spectrum with a 2-GHz guard band. Linear pre-emphasis is employed to compensate for transmitter bandwidth limitations.

At the receiver, the waveform is first re-sampled to 2 samples-per-symbol (SPSs). The frequency offset estimation (FOE) is realized by searching the peak of the signal spectrum (i.e., the pilot tone). Carrier phase recovery (CPR) is subsequently performed by a digital low-pass filter (LPF) to extract the phase trace on pilot tone [9]. After matched RRC filtering, frame synchronization is implemented by cross-correlation based on preamble. We then conduct T/2-spaced linear multi-input-multi-output (MIMO) equalization updated by the recursive least square (RLS) algorithm. After down-sampling to 1 SPS, blind

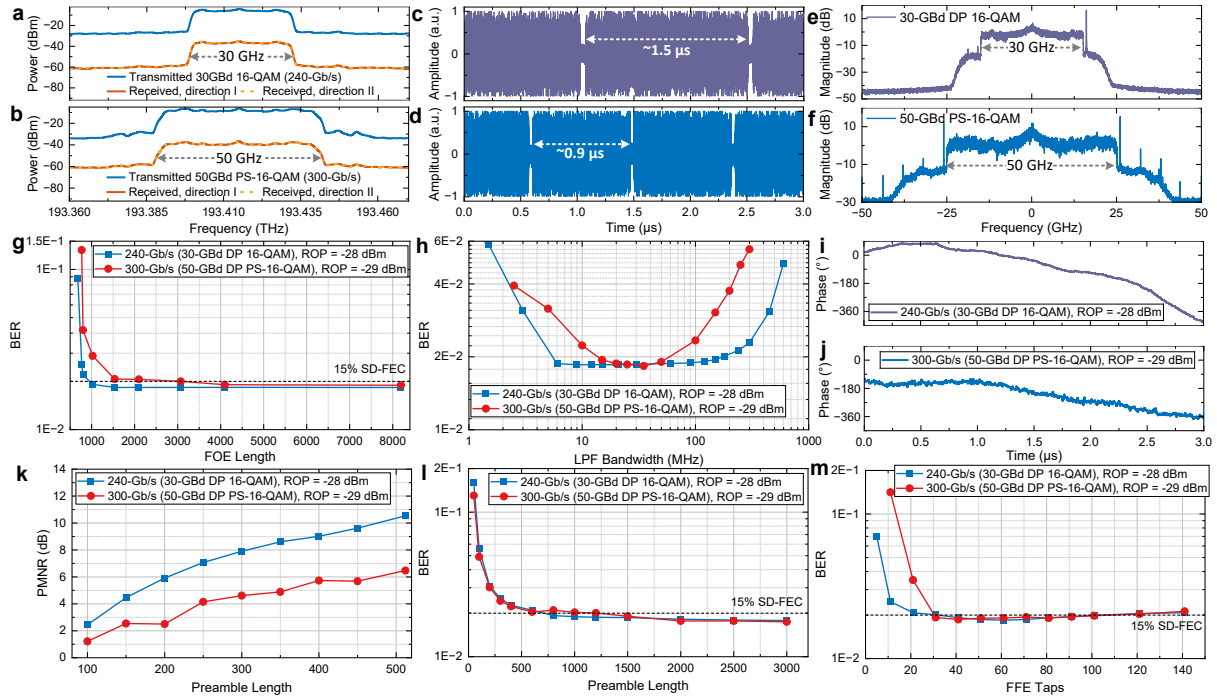


Fig. 2: Measured optical spectra with (a) 30-GBd and (b) 50-GBd signals. Received electrical waveforms of (c) 30-GBd and (d) 50-GBd signals. Received electrical spectra of (e) 30-GBd and (f) 50-GBd signals. (g) Measured BER vs. sequence length for FOE stage. (h) Measured BER vs. low-pass filter (LPF) bandwidth for CPR stage. Estimated phase noise of (i) 30-GBd and (j) 50-GBd signals. (k) Measured peak-to-maximum noise ratio (PMNR) vs. preamble length for frame synchronization. (l) Measured BER vs. preamble length for channel equalization. (m) Measured BER vs. FFE tap length for channel equalization.

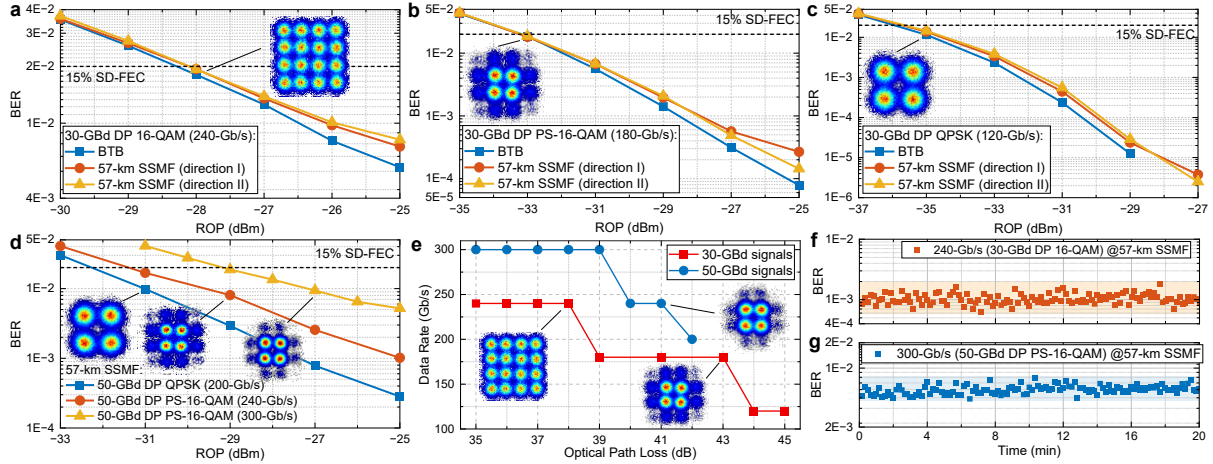


Fig. 3: Measured BER versus ROP with (a) 240-Gb/s DP 16-QAM, (b) 180-Gb/s DP PS-16-QAM, and (c) 120-Gb/s DP QPSK signals, respectively. (d) Measured BER versus ROP with 50-GBd signals from 200 to 300 Gb/s after 57-km field-deployed fiber transmission. (e) Measured data rate versus optical path loss (OPL). Long-term BER stability test with (f) 240-Gb/s DP 16-QAM, and (g) 300-Gb/s DP PS-16-QAM signals, respectively.

phase search is used to suppress the residual phase rotation within $\pm 6^\circ$.

Experimental Results and Discussions

Fig. 2(a) and 2(b) show the transmitted and received optical spectra of 30- and 50-GBd signals. With pre-emphasis, the spectra are nearly flat, and similar losses are observed in both transmission directions. Fig. 2(c) and 2(d) depict the received electrical waveforms, with frame durations of 1.5 and 0.9 μ s for 30- and 50-GBd signals, respectively. Fig. 2(e) and 2(f) present the electrical spectra. For 30-GBd reception, a 20-GHz low-pass filter is applied to suppress out-of-band noise, while the 50-GBd signal exhibits noticeable I/Q imbalance, leading to a mirrored pilot tone.

We then optimize the fast-convergence DSP. Fig. 2(g) shows the measured BER versus sequence length for the FOE stage, indicating that 3000 symbols are sufficient to reliably identify the pilot tone. As FOE operates on blind symbols, it is not included in the preamble overhead. Fig. 2(h) presents the BER versus LPF bandwidth for the CPR stage, where an optimal value of 30 MHz is obtained by balancing additive noise and phase noise. Fig. 2(i) and 2(j) show the estimated phase noise traces for 30- and 50-GBd signals, with increased fluctuations at 50 GBd due to lower SNR (-29 dBm ROP).

Fig. 2(k) evaluates frame synchronization performance versus preamble length using the peak-to-maximum noise ratio (PMNR). As shown in Fig. 2(l), 600 symbols enable initial convergence, while 1500 symbols reduce the BER below the 15% SD-FEC threshold of 2×10^{-2} . Fig. 2(m) shows the BER versus tap length for the channel equalization stage. Less than 20 taps cannot fully mitigate inter-symbol interference (ISI), whereas excessive taps introduce noise accumulation. An optimal tap number of 41 is selected.

The ROP sensitivity with flexible-rate operation

is further evaluated. Fig. 3(a)–3(c) show BER versus ROP for DP 30-GBd signals using 16-QAM, PS-16-QAM (entropy = 3 bit/symbol), and QPSK formats, achieving line rates of 240, 180, and 120 Gb/s, respectively. At the 15% SD-FEC threshold, the sensitivities are -28 , -33 , and -35 dBm, corresponding to power budgets of 38, 43, and 45 dB with a 10-dBm launch power. Compared with the back-to-back (BTB) case, negligible penalty is observed after 57-km field transmission. We attribute this phenomenon to the high connection loss in Site A and B, which reduces the launch power when feeding into the field fiber link.

We boost the symbol rate to 50 GBd and evaluate the sensitivity. We employ QPSK, PS-16-QAM (entropy = 2.4 bit/symbol), and PS-16-QAM (entropy = 3 bit/symbol) to reach line rates of 200, 240, and 300 Gb/s, respectively. The ROP sensitivity after 57-km SSMF transmission are -32 , -31 , and -29 dBm, corresponding to power budgets of 42, 41, and 39 dB. Fig. 3(e) summarizes the data rates and power budgets. It is found that lower modulation formats provide superior noise tolerance and can achieve higher budgets. So, it is beneficial to increase the baud rate, if transceiver bandwidth is supported. Moreover, we test the stability of the DSP with 240- and 300-Gb/s signals in Fig. 2(f) and 2(g). The BERs remain well below the 15% SD-FEC threshold for 20 minutes. Long-term stability can be potentially extended if measuring time is allowed.

Conclusions

We present the first field trial of 200/300-Gb/s coherent TDM-PON over 57-km field-installed fiber with 42/39-dB power budget. The preamble length can be optimized to 30 ns with fast-convergence and stable DSP. This field trial demonstration verifies the feasibility of next-generation coherent PON with higher data rates and extended reach.

Acknowledgements

This work was supported by National Key Research and Development Program of China (2024YFB2908005, 2023YFB2906000), and National Natural Science Foundation of China (62475147, 62271305).

References

- [1] ITU-T, "Higher speed passive optical networks," ITU-T Recommendation G.9804 series, 2025.
- [2] D. Zhang, D. Lu, and D. Nessel, "Progress of very high speed PON in ITU-T," *Journal of Optical Communications and Networking*, vol. 18, no. 2, pp. A112-A121, 2026. DOI: [10.1364/JOCN.573747](https://doi.org/10.1364/JOCN.573747)
- [3] G. Rizzelli and R. Gaudino, "High-speed PON solutions: Recent evolutions and expected trends," *Journal of Optical Communications and Networking*, vol. 18, no. 2, pp. A134-A149, 2026. DOI: [10.1364/JOCN.575601](https://doi.org/10.1364/JOCN.575601)
- [4] M. S. Faruk, X. Li, D. Nessel, I. N. Cano, A. Rafel, and S. J. Savory, "Coherent passive optical networks: Why, when, and how," *IEEE Communications Magazine*, vol. 59, no. 12, pp. 112-117, 2021. DOI: [10.1109/MCOM.010.2100503](https://doi.org/10.1109/MCOM.010.2100503)
- [5] Y. Zhu, L. Yi, B. Yang, X. Huang, J. S. Wey, Z. Ma, W. Hu, "Comparative study of cost-effective coherent and direct detection schemes for 100 Gb/s/λ PON," *Journal of Optical Communications and Networking*, vol. 12, no. 9, pp. D36-D47, 2020. DOI: [10.1364/JOCN.573289](https://doi.org/10.1364/JOCN.573289)
- [6] G. Yang, Y. Zhu, X. Fang, Z. Zhang, L. Man, W. Hu, F. Zhang, and Z. Ma, "Hardware- and DSP-efficient 200-Gb/s coherent PON insensitive to frequency offset and phase noise for burst mode upstream," *Proc. ECOC'2024*, paper W2A.127.
- [7] A. Yan, R. Zheng, H. Xu, P. Luo, Z. Zhao, M. Shen, S. He, S. Xing, Y. Hu, X. Deng, J. Zhao, B. Dong, Y. Liu, O. Huang, D. Zhang, L. Zhang, Y. Zhou, D. Guo, J. Zhou, N. Chi, and J. Zhang, "World's first field-trial demonstration of bidirectional 200G TFDMA coherent PON enabled by real-time FPGA-based reception," *Proc. OFC'2026*, paper Th4C.4.
- [8] I. B. Kovacs, M. S. Faruk, A. Wonfor, and S. J. Savory, "200 Gb/s/λ bidirectional coherent PON solutions demonstrated over field installed fiber," *IEEE Photonics Technology Letters*, vol. 37, no. 5, pp. 257-260, 2025. DOI: [10.1109/LPT.2024.3434640](https://doi.org/10.1109/LPT.2024.3434640)
- [9] Y. Zhu, X. Huang, X. Fang, X. Huang, Z. Zhang, L. Man, G. Li, Y. Hu, Y. Zhong, F. Zhang, and W. Hu, "200G coherent TFDMA PON with flexible subcarrier reception based on a low-cost ONU," *Journal of Optical Communications and Networking*, vol. 18, no. 2, pp. A100-A111, 2026. DOI: [10.1364/JOCN.573289](https://doi.org/10.1364/JOCN.573289)